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PHY211



Covenant University
Raising a new Generation of Leaders

ELEMENTARY MODERN PHYSICS

BY:

BABARIMISA O. IDOWU

OFFICE NO: ROOM 187, LECTURE THEATRE

THE ATOM DEFINITION

- An atom is the “smallest” particle of any element that still retains the characteristics of that element.

- A piece of an element that we are able to see or handle is made of many atoms and all atoms of a particular element are the same…

  they all have the same number of protons.

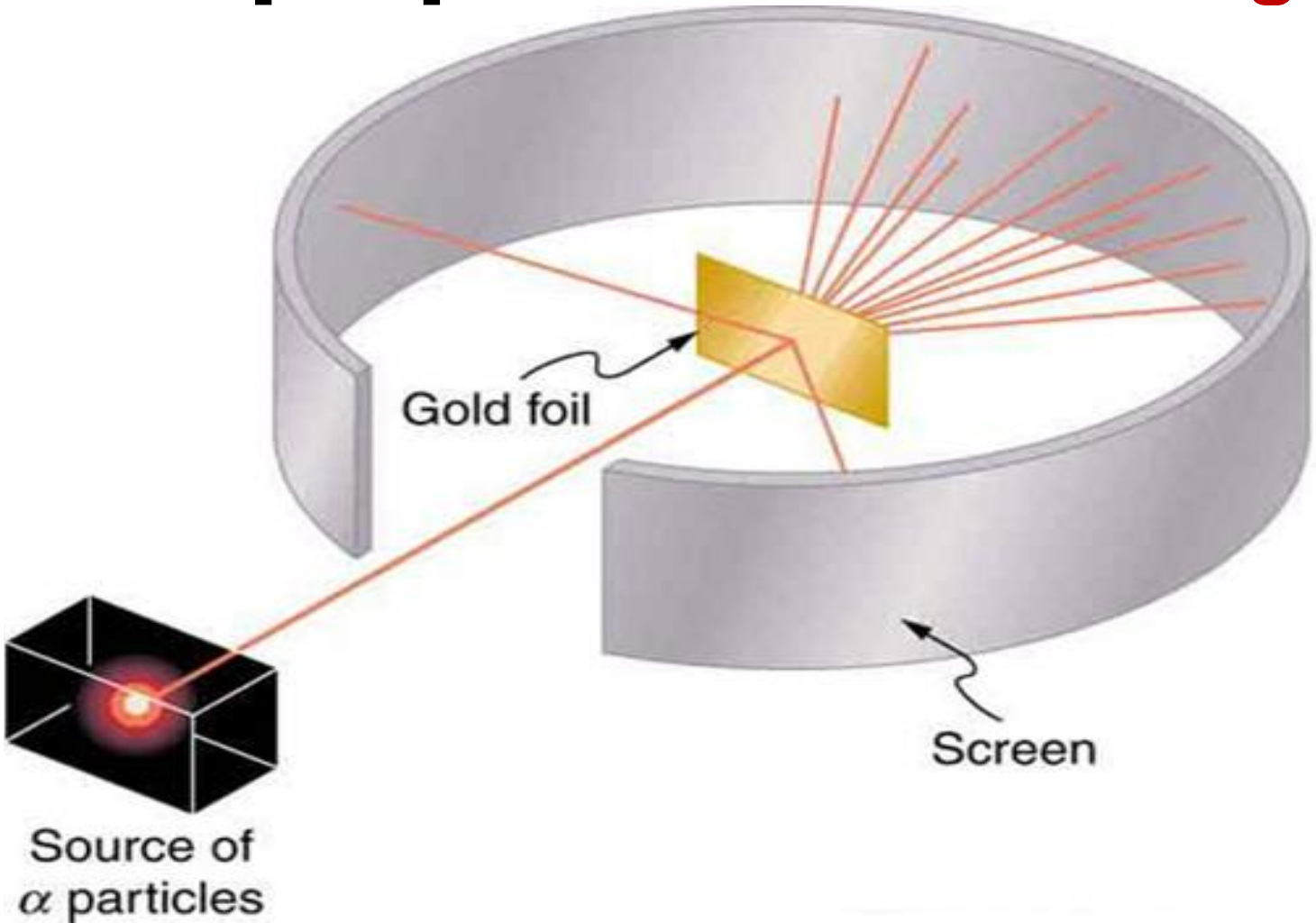
THE ATOMIC STRUCTURE

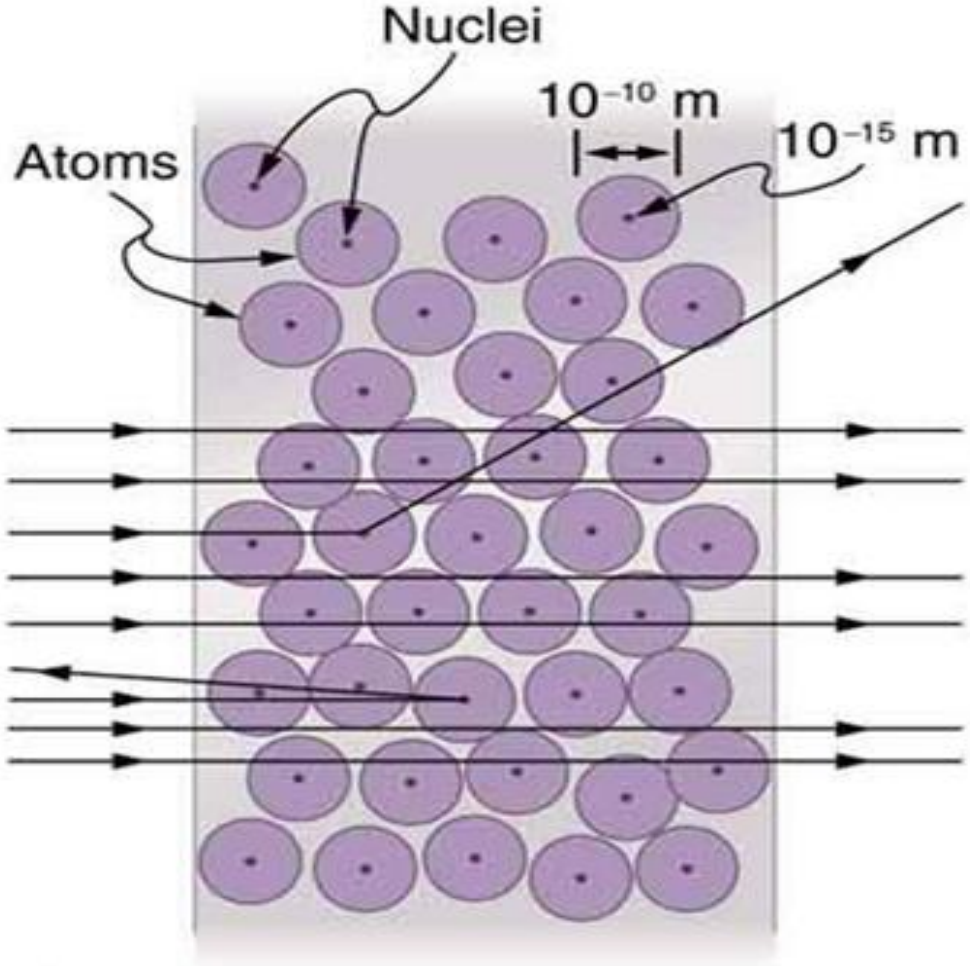


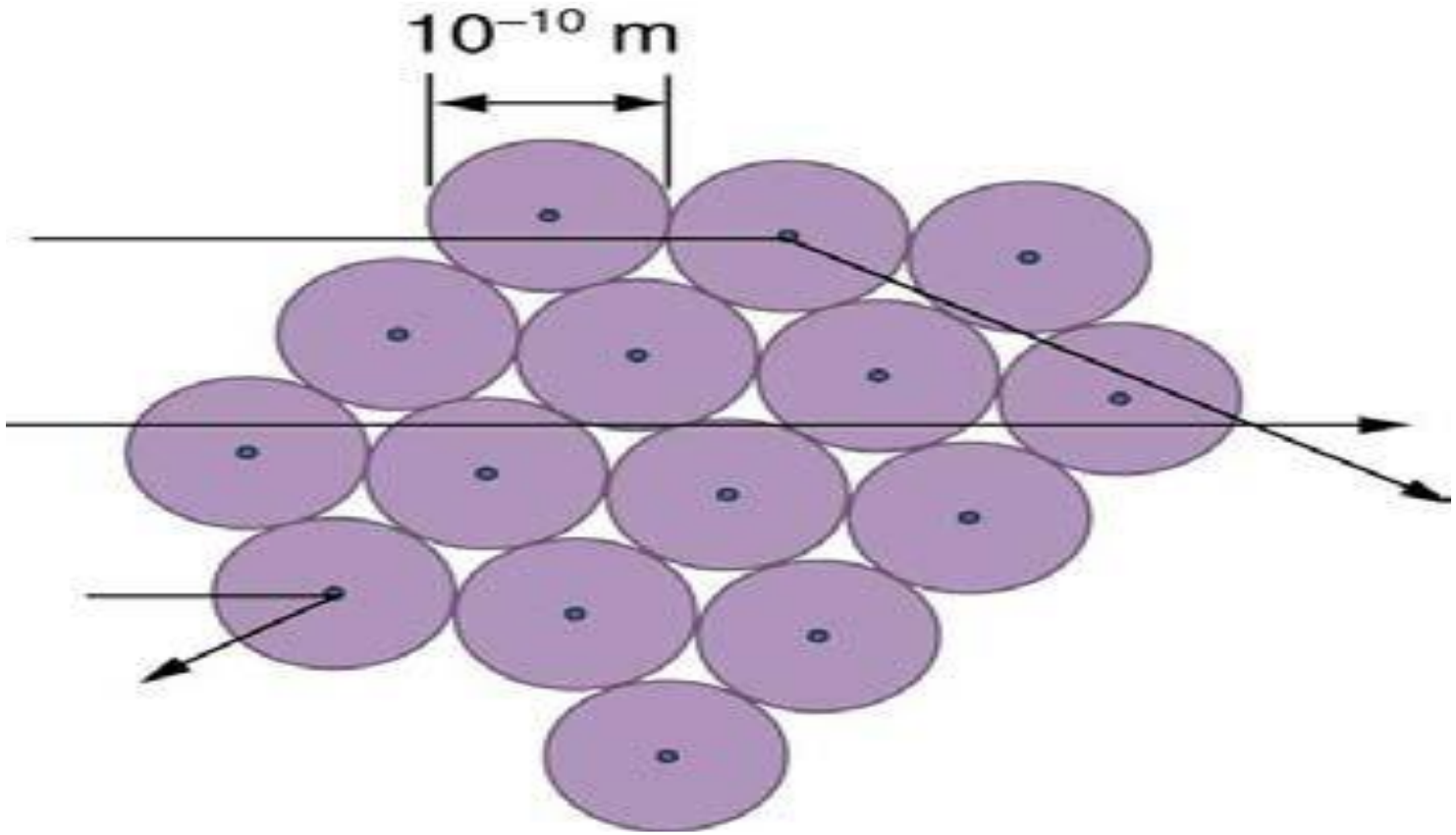
- **The emission of light from atoms is an important clue that allows us to learn about the structure of the atom.**
- **The colors are determined by the types of atoms in the material burning in the explosion:**
- **Bright white light often comes from oxidizing magnesium or aluminum.**
- **Red light often comes from strontium**
- **Yellow from sodium.**
- **Blue light is more difficult to achieve, but can be obtained by burning a mixture of copper powder, copper chloride, and hexachloroethane.**

Rutherford's experiment

- **Rutherford's experiment gave direct evidence for the size and mass of the nucleus by scattering alpha particles from a thin gold foil.**





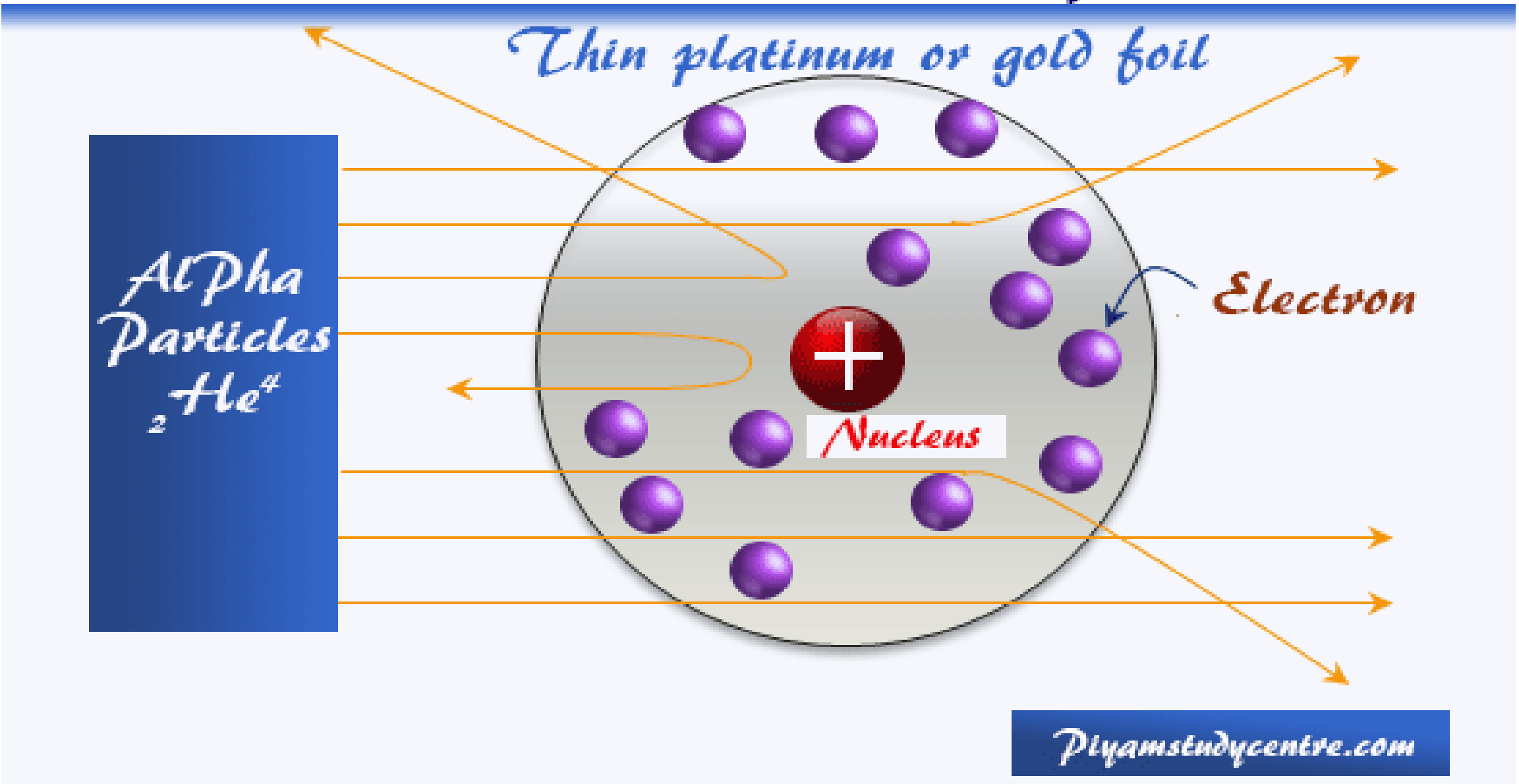


Expanded view of the atoms in the gold foil in Rutherford's experiment.

Circles represent the atoms (about 10^{-10} m in diameter), while the dots represent the nuclei (about 10^{-15} m in diameter)

To be visible, the dots are much larger than scale.

Rutherford Gold Foil Experiment



CREDIT-https://www.priyamstudycentre.com

OBSERVATIONS:

- Most alpha particles crash through and are relatively unaffected because of their high energy and the electron's small mass.
- Some, however, head straight toward a nucleus and are scattered straight back.
- A detailed analysis gives the size and mass of the nucleus.

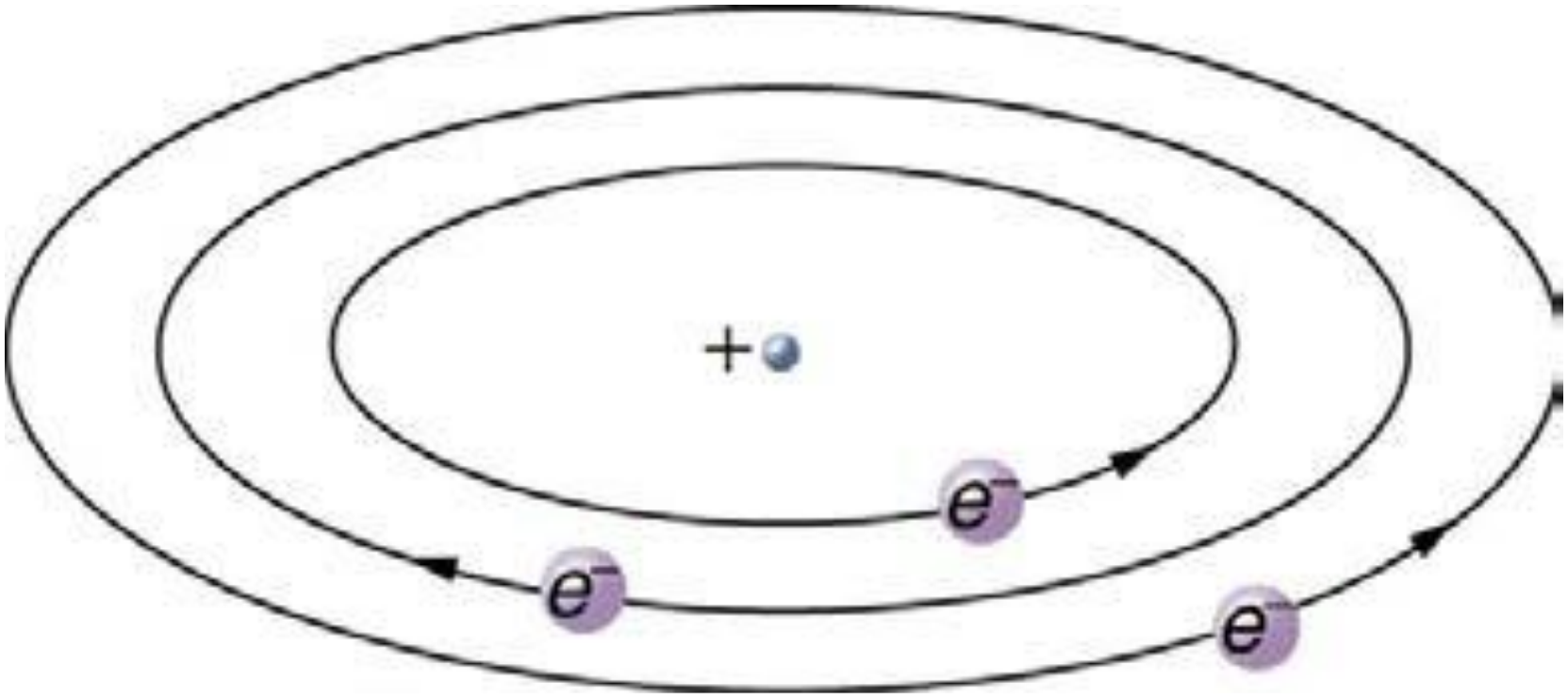
- Alpha particles (doubly charged positive nuclei of helium atoms) with energies of about 5 MeV are emitted from a radioactive source (which is a small metal container in which a specific amount of a radioactive material is sealed), are collimated into a beam, and fall upon the thin gold foil.
- The number of particles that penetrate the foil or scatter to various angles indicates that gold nuclei are very small and contain nearly all of the mass of the gold.
- This is particularly indicated by the alpha particles that scatter to very large angles, much like a soccer ball bouncing off a goalie's head.

- **Based on the size and mass of the nucleus revealed by his experiment, as well as the mass of electrons, Rutherford proposed the planetary model of the atom.**

Rutherford's PLANETARY MODEL OF THE ATOM

- The model incorporates the characteristics of the nucleus, electrons, and the size of the atom.

First to recognize the structure of atoms, in which low-mass electrons orbit a very small, massive nucleus in orbits much larger than the nucleus.



- Rutherford's planetary model of the atom was crucial to understanding the characteristics of atoms, and their interactions and energies.
- It was an indication of how different nature is from the familiar classical world on the small, quantum mechanical scale.
- Search has continued for deeper substructures, such as those inside the nucleus, with some success in discovering quarks and other elementary particles.

Bohr's Theory of the Hydrogen Atom

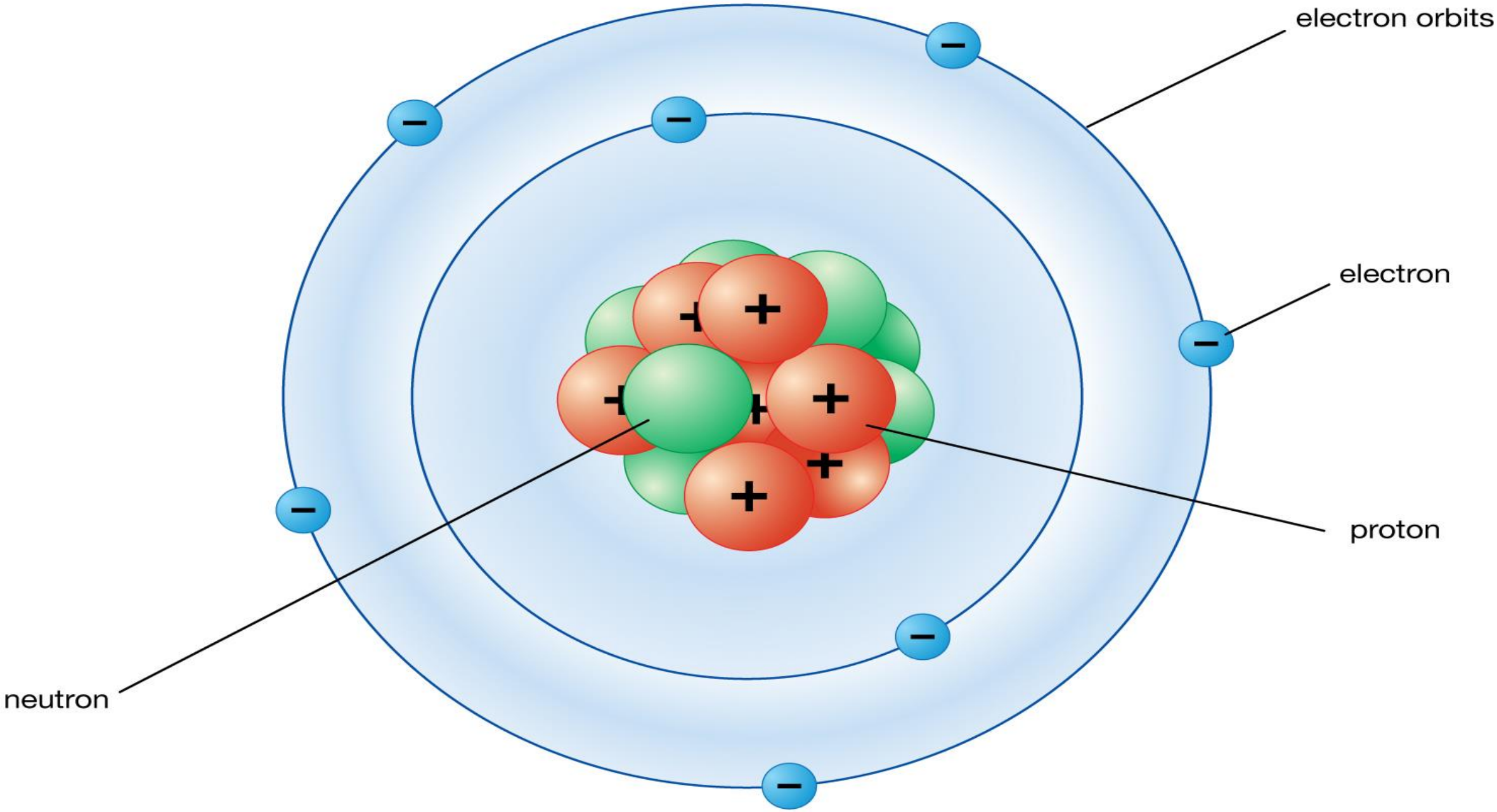
Definition:

- A theory of atomic structure in which the hydrogen atom (Bohr atom,) is assumed to consist of a proton as nucleus, with a single electron moving in distinct circular orbits around it, each orbit corresponding to a specific quantized energy state.

- Bohr's theory explained the atomic spectrum of hydrogen and established new and broadly applicable principles in quantum mechanics.

- In 1913, Niels Bohr created a picture of the atom that was called the Bohr Model.
- Bohr used the planetary model of the atom to explain the atomic spectrum and size of the hydrogen atom
- Bohr proposed that electrons revolve around the nucleus.
- Within each orbit, each electron has a fixed amount of energy in it.

Bohr atomic model of a nitrogen atom



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- **The Bohr model of the hydrogen atom (Z = 1)**
- **or a hydrogen-like ion (Z > 1),**

where the negatively-charged electron, confined to an atomic shell encircles a small, positively charged atomic nucleus

and when an electron jumps between orbits;

it is accompanied by an emitted or absorbed amount of electromagnetic energy (*hv*)

The Bohr model works only for hydrogen because it considers only the interactions between one electron and the nucleus.

The Bohr model is based on the energy levels of one electron orbiting a nucleus at various energy levels.

Any other electrons in the atom will repel the one electron and change its energy level.

What are the two limitations of the Bohr model for the atom?

1. **It is in violation of the Heisenberg Uncertainty Principle**

   ***The Heisenberg uncertainty principle states that it is impossible to know simultaneously the exact position and momentum of a particle*.**

   The Bohr Model considers electrons to have both a known radius and orbit, which is impossible according to Heisenberg.

2. **The Bohr Model is very limited in terms of size.**

   Poor spectral predictions are obtained when larger atoms are considered.

Schrodinger's atomic model:

- In 1926 Erwin Schrödinger, an Austrian physicist, took the Bohr atom model one step further.

Schrödinger used mathematical equations to describe the likelihood / probability of finding an electron in a certain position.

This atomic model is known as the Quantum Mechanical Model of the Atom.

Definition of Quantum Mechanical Model or Wave Mechanical Model of the atom:

The atom is mostly space with a small, very dense, centrally located nucleus that contains positively charged protons and neutrally charged neutrons and is surrounded by negatively charged electrons located in orbitals in the electron cloud.

(*See the video*)

How does the quantum mechanical model of the atom describe the electron structure of the atom?

- **Bohr proposed that an electron is found only in specific circular paths, or orbits, around the nucleus.**
  Each possible electron orbit in Bohr's model has a fixed energy.

- **What does the quantum mechanical model determine about the electron in the atom? … The orbitals describe where an electron is likely to be found.**

The quantum mechanical model of the atom due to the Uncertainty Principle:

- **The quantum mechanical model of the atom uses complex shapes of orbitals (sometimes called electron clouds), volumes of space in which there is likely to be an electron.**
  So, this model is based on probability rather than certainty.

Basis for the quantum mechanical model of the atom:

- **In the Quantum Mechanical Model of the Atom, Energy Is Quantized.**

  After Max Planck determined that energy is released and absorbed by atoms in certain fixed amounts known as quanta;
  Albert Einstein took his work a step further, determining that radiant energy is also quantized;

  He called the discrete energy packets *photons*.

What is the difference between the Bohr model and the Quantum Mechanical Model?

- **In the Bohr Model, the electron is treated as a particle in fixed orbits around the nucleus.**
- **In the Quantum Mechanical Model, the electron is treated mathematically as a wave.**

❖ ***The electron has properties of both particles and waves.***

❖ Most modern model of the atom:

Electrons are in **constant motion around** the **nucleus;**

protons and **neutrons** jiggle within the **nucleus,**

and

quarks jiggle within the **protons** and **neutrons**